

XPS Database - References

- 1.** Laure Caracciolo; Lénaïc Madec; Hervé Martinez. XPS Analysis of K-based Reference Compounds to Allow Reliable Studies of Solid Electrolyte Interphase in K-ion Batteries. *ACS Applied Energy Materials*, 2021, Vol. 4, 11693-11699. DOI: 10.1021/acsaem.1c02400
- 2.** R.I.R Blyth; H Buqa; F.P Netzer; M.G Ramsey; J.O Besenhard; P Golob; M Winter. XPS studies of graphite electrode materials for lithium ion batteries. *Applied Surface Science*, 2000, Vol. 167, 99-106. DOI: 10.1016/S0169-4332(00)00525-0
- 3.** R. Dedryvère; L. Gireaud; S. Grugeon; S. Laruelle; J.-M. Tarascon; D. Gonbeau. Characterization of Lithium Alkyl Carbonates by X-ray Photoelectron Spectroscopy: Experimental and Theoretical Study. *The Journal of Physical Chemistry B*, 2005, Vol. 109, 15868-15875. DOI: 10.1021/jp051626k
- 4.** L. El Ouatani; R. Dedryvère; C. Siret; P. Biensan; S. Reynaud; P. Iratçabal; D. Gonbeau. The Effect of Vinylene Carbonate Additive on Surface Film Formation on Both Electrodes in Li-Ion Batteries. *Journal of The Electrochemical Society*, 2009, Vol. 156, A103. DOI: 10.1149/1.3029674
- 5.** A. Shchukarev; D. Korolkov. XPS Study of group IA carbonates. *Central European Journal of Chemistry*, 2004, Vol. 2, 347-362. DOI: 10.2478/BF02475578
- 6.** R. Dedryvère; D. Foix; S. Franger; S. Patoux; L. Daniel; D. Gonbeau. Electrode/Electrolyte Interface Reactivity in High-Voltage Spinel $\text{LiMn}_{1.6}\text{Ni}_{0.4}\text{O}_4/\text{Li}_4\text{Ti}_5\text{O}_{12}$ Lithium-Ion Battery. *The Journal of Physical Chemistry C*, 2010, Vol. 114, 10999-11008. DOI: 10.1021/jp1026509
- 7.** Hitoshi Ota; Toshio Akai; Hideo Namita; Shoji Yamaguchi; Masaharu Nomura. XAFS and TOF-SIMS analysis of SEI layers on electrodes. *Journal of Power Sources*, 2003, Vol. 119-121, 567-571. DOI: 10.1016/S0378-7753(03)00291-X
- 8.** Bertrand Philippe; Rémi Dedryvère; Mihaela Gorgoi; Håkan Rensmo; Danielle Gonbeau; Kristina Edström. Improved Performances of Nanosilicon Electrodes Using the Salt LiFSI: A Photoelectron Spectroscopy Study. *Journal of the American Chemical Society*, 2013, Vol. 135, 9829-9842. DOI: 10.1021/ja403082s
- 9.** Bertrand Philippe; Rémi Dedryvère; Joachim Allouche; Fredrik Lindgren; Mihaela Gorgoi; Håkan Rensmo; Danielle Gonbeau; Kristina Edström. Nanosilicon Electrodes for Lithium-Ion Batteries: Interfacial Mechanisms Studied by Hard and Soft X-ray Photoelectron Spectroscopy. *Chemistry of Materials*, 2012, Vol. 24, 1107-1115. DOI: 10.1021/cm2034195
- 10.** F. J. Himpsel; F. R. McFeely; A. Taleb-Ibrahimi; J. A. Yarmoff; G. Hollinger. Microscopic structure of the SiO_2/Si interface. *Physical Review B*, 1988, Vol. 38, 6084-6096. DOI: 10.1103/PhysRevB.38.6084
- 11.** Mark C. Biesinger; Brad P. Payne; Leo W. M. Lau; Andrea Gerson; Roger St. C. Smart. X-ray photoelectron spectroscopic chemical state quantification of mixed nickel metal, oxide and hydroxide systems. *Surface and Interface Analysis*, 2009, Vol. 41, 324-332. DOI: 10.1002/sia.3026
- 12.** Andrew P. Grosvenor; Mark C. Biesinger; Roger St.C. Smart; N. Stewart McIntyre. New interpretations of XPS spectra of nickel metal and oxides. *Surface Science*, 2006, Vol. 600, 1771-1779. DOI: 10.1016/j.susc.2006.01.041
- 13.** Mark C. Biesinger; Leo W. M. Lau; Andrea R. Gerson; Roger St. C. Smart. The role of the Auger parameter in XPS studies of nickel metal, halides and oxides. *Physical Chemistry Chemical Physics*, 2012, Vol. 14, 2434. DOI: 10.1039/c2cp22419d
- 14.** Mark C. Biesinger; Brad P. Payne; Andrew P. Grosvenor; Leo W.M. Lau; Andrea R. Gerson; Roger St.C. Smart. Resolving surface chemical states in XPS analysis of first row transition metals, oxides and hydroxides: Cr, Mn, Fe, Co and Ni. *Applied Surface Science*, 2011, Vol. 257, 2717-2730. DOI: 10.1016/j.apsusc.2010.10.051

- 15.** S. Cotte; B. Pecquenard; F. Le Cras; R. Grissa; H. Martinez; L. Bourgeois. Lithium-rich manganese oxide spinel thin films as 3 V electrode for lithium batteries. *Electrochimica Acta*, 2015, Vol. 180, 528-534. DOI: 10.1016/j.electacta.2015.08.145
- 16.** H. Bouayad; Z. Wang; N. Dupre; R. Dedryvere; D. Foix; S. Franger; J.-F. Martin; L. Boutafa; S. Patoux; D. Gonbeau; D. Guyomard. Improvement of Electrode/Electrolyte Interfaces in High-Voltage Spinel Lithium-Ion Batteries by Using Glutaric Anhydride as Electrolyte Additive. *The Journal of Physical Chemistry C*, 2014, Vol. 118, 4634-4648. DOI: 10.1021/jp5001573
- 17.** R. Grissa; H. Martinez; S. Cotte; J. Galipaud; B. Pecquenard; F. Le Cras. Thorough XPS analyses on overlithiated manganese spinel cycled around the 3 V plateau. *Applied Surface Science*, 2017, Vol. 411, 449-456. DOI: 10.1016/j.apsusc.2017.03.205
- 18.** R. Grissa; H. Martinez; V. Pele; S. Cotte; B. Pecquenard; F. Le Cras. An X-ray photoelectron spectroscopy study of the electrochemical behaviour of iron molybdate thin films in lithium and sodium cells. *Journal of Power Sources*, 2017, Vol. 342, 796-807. DOI: 10.1016/j.jpowsour.2016.12.117
- 19.** S. Cotte; V. Pele; B. Pecquenard; F. Le Cras; R. Grissa; L. Bourgeois; M.T. Sougrati; H. Martinez. Iron molybdate thin films prepared by sputtering and their electrochemical behavior in Li batteries. *Journal of Alloys and Compounds*, 2018, Vol. 735, 1454-1462. DOI: 10.1016/j.jallcom.2017.11.231
- 20.** L. Bodenes; A. Darwiche; L. Monconduit; H. Martinez. The Solid Electrolyte Interphase a key parameter of the high performance of Sb in sodium-ion batteries: Comparative X-ray Photoelectron Spectroscopy study of Sb/Na-ion and Sb/Li-ion batteries. *Journal of Power Sources*, 2015, Vol. 273, 14-24. DOI: 10.1016/j.jpowsour.2014.09.037
- 21.** Laure Caracciolo; L  na  c Madec; Emmanuel Petit; Vincent Gabaudan; Dany Carlier; Laurence Croguennec; Herv   Martinez. Electrochemical Redox Processes Involved in Carbon-Coated KVPO4F for High Voltage K-Ion Batteries Revealed by XPS Analysis. *Journal of The Electrochemical Society*, 2020, Vol. 167, 130527. DOI: 10.1149/1945-7111/abbb0c
- 22.** Leonie Wildersinn; Fabian Jeschull. The influence of sodium impurities in metallic potassium on photoemission spectra and electrochemical data. *RSC Applied Interfaces*, 2026. DOI: 10.1039/D6LF00207B
- 23.** Leonie Wildersinn; Daniel Stottmeister; Fabian Jeschull; Axel Gro  ; Andreas Hofmann. Decomposition of Binary Mixtures of DMC/EC, EMC/EC, and DEC/EC on Potassium Surfaces; GC, XPS, and Calculation. *ACS Applied Materials & Interfaces*, 2025, Vol. 17, 10055-10072. DOI: 10.1021/acsami.4c17461
- 24.** Geert Silversmit; Diederik Depla; Hilde Poelman; Guy B. Marin; Roger De Gryse. Determination of the V2p XPS binding energies for different vanadium oxidation states (V5+ to V0+). *Journal of Electron Spectroscopy and Related Phenomena*, 2004, Vol. 135, 167-175. DOI: 10.1016/j.elspec.2004.03.004
- 25.** E. Hryha; E. Rutqvist; L. Nyborg. Stoichiometric vanadium oxides studied by XPS. *Surface and Interface Analysis*, 2012, Vol. 44, 1022-1025. DOI: 10.1002/sia.3844
- 26.** A. Benayad; H. Martinez; A. Gies; B. Pecquenard; A. Levasseur; D. Gonbeau. XPS investigations achieved on the first cycle of V2O5 thin films used in lithium microbatteries. *Journal of Electron Spectroscopy and Related Phenomena*, 2006, Vol. 150, 1-10. DOI: 10.1016/j.elspec.2005.09.001
- 27.** Eugene S. Ilton; Jeffrey E. Post; Peter J. Heaney; Florence T. Ling; Sebastien N. Kerisit. XPS determination of Mn oxidation states in Mn (hydr)oxides. *Applied Surface Science*, 2016, Vol. 366, 475-485. DOI: 10.1016/j.apsusc.2015.12.159
- 28.** Yisi Zhu; Justin G. Connell; Sanja Tepavcevic; Peter Zapol; Regina Garcia-Mendez; Nathan J. Taylor; Jeff Sakamoto; Brian J. Ingram; Larry A. Curtiss; John W. Freeland; Dillon D. Fong; Nenad M.

Markovic. Dopant-Dependent Stability of Garnet Solid Electrolyte Interfaces with Lithium Metal. *Advanced Energy Materials*, 2019, Vol. 9, 1803440. DOI: 10.1002/aenm.201803440

29. Jérémie Auvergniot; Alice Cassel; Dominique Foix; Virginie Viallet; Vincent Seznec; Rémi Dedryvère. Redox activity of argyrodite $\text{Li}_6\text{PS}_5\text{Cl}$ electrolyte in all-solid-state Li-ion battery: An XPS study. *Solid State Ionics*, 2017, Vol. 300, 78-85. DOI: 10.1016/j.ssi.2016.11.029

30. Marzia Fantauzzi; Bernhard Elsener; Davide Atzei; Americo Rigoldi; Antonella Rossi. Exploiting XPS for the identification of sulfides and polysulfides. *RSC Advances*, 2015, Vol. 5, 75953-75963. DOI: 10.1039/C5RA14915K

31. Xin Zhang; Haoyu Yang; Yujiang Sun; Yongan Yang. Lithium Sulfide: Magnesothermal Synthesis and Battery Applications. *ACS Applied Materials & Interfaces*, 2022, Vol. 14, 41003-41012. DOI: 10.1021/acsami.2c11196

32. N. Andreu; D. Flahaut; R. Dedryvère; M. Minvielle; H. Martinez; D. Gonbeau. XPS Investigation of Surface Reactivity of Electrode Materials: Effect of the Transition Metal. *ACS Applied Materials & Interfaces*, 2015, Vol. 7, 6629-6636. DOI: 10.1021/am5089764

33. R. Dedryvère; S. Laruelle; S. Grugeon; P. Poizot; D. Gonbeau; J.-M. Tarascon. Contribution of X-ray Photoelectron Spectroscopy to the Study of the Electrochemical Reactivity of CoO toward Lithium. *Chemistry of Materials*, 2004, Vol. 16, 1056-1061. DOI: 10.1021/cm0311269

34. C.S. Nimisha; K. Yellareswar Rao; G. Venkatesh; G. Mohan Rao; N. Munichandraiah. Sputter deposited LiPON thin films from powder target as electrolyte for thin film battery applications. *Thin Solid Films*, 2011, Vol. 519, 3401-3406. DOI: 10.1016/j.tsf.2011.01.087

35. Sebastian Wenzel; Thomas Leichtweiß; Dominik A. Weber; Joachim Sann; Wolfgang G. Zeier; Jürgen Janek. Interfacial Reactivity Benchmarking of the Sodium Ion Conductors Na_3PS_4 and Sodium β -Alumina for Protected Sodium Metal Anodes and Sodium All-Solid-State Batteries. *ACS Applied Materials & Interfaces*, 2016, Vol. 8, 28216-28224. DOI: 10.1021/acsami.6b10119

36. Palanivel Molaiyan; Sarah E. Mailhot; Kevin Voges; Anu M. Kantola; Tao Hu; Peter Michalowski; Arno Kwade; Ville-Veikko Telkki; Ulla Lassi. Investigation of the structure and ionic conductivity of a Li_3InCl_6 modified by dry room annealing for solid-state Li-ion battery applications. *Materials & Design*, 2023, Vol. 227, 111690. DOI: 10.1016/j.matdes.2023.111690

37. Luise M. Riegger; Roman Schlem; Joachim Sann; Wolfgang G. Zeier; Jürgen Janek. Lithium-Metal Anode Instability of the Superionic Halide Solid Electrolytes and the Implications for Solid-State Batteries. *Angewandte Chemie International Edition*, 2021, Vol. 60, 6718-6723. DOI: 10.1002/anie.202015238

38. Juntao Shi; Zhujun Yao; Jiayuan Xiang; Chen Cai; Fangfang Tu; Yongqi Zhang; Weilin Yao; Qixiang Jia; Yan Zhou; Shenghui Shen; Yefeng Yang. High-Conductivity Li_2ZrCl_6 Electrolytes via an Optimized Two-Step Ball-Milling Method for All-Solid-State Lithium-Metal Batteries. *ACS Sustainable Chemistry & Engineering*, 2024, Vol. 12, 2009-2017. DOI: 10.1021/acssuschemeng.3c06652

39. Sebastian Wenzel; Dominik A. Weber; Thomas Leichtweiss; Martin R. Busche; Joachim Sann; Jürgen Janek. Interphase formation and degradation of charge transfer kinetics between a lithium metal anode and highly crystalline $\text{Li}_7\text{P}_3\text{S}_{11}$ solid electrolyte. *Solid State Ionics*, 2016, Vol. 286, 24-33. DOI: 10.1016/j.ssi.2015.11.034

40. Yiqun Wang; Takuya Matsuyama; Minako Deguchi; Akitoshi Hayashi; Aiko Nakao; Masahiro Tatsumisago. X-ray photoelectron spectroscopy for sulfide glass electrolytes in the systems Li_2S - P_2S_5 and Li_2S - P_2S_5 - LiBr . *Journal of the Ceramic Society of Japan*, 2016, Vol. 124, 597-601. DOI: 10.2109/jcersj2.16006

41. Guanghua Yang; Junli Shi; Cai Shen; Shuwei Wang; Lan Xia; Huasheng Hu; Hao Luo; Yonggao Xia; Zhaoping Liu. Improving the cyclability performance of lithium-ion batteries by introducing lithium difluorophosphate (LiPO_2F_2) additive. *RSC Advances*, 2017, Vol. 7, 26052-26059. DOI: 10.1039/C7RA03926C

- 42.** B.V.R. Chowdari; K.L. Tan; W.T. Chia. Raman and X-ray photoelectron spectroscopic studies of lithium phosphotungstate glasses. *Solid State Ionics*, 1992, Vol. 53-56, 1172-1178. DOI: 10.1016/0167-2738(92)90308-C
- 43.** Emil N. Rizkalla. X-ray Photoelectron Studies of some Phosphates. *Inorganica Chimica Acta*, 1982, Vol. 60, 53-57. DOI: 10.1016/S0020-1693(00)91150-2
- 44.** Roozbeh Hessabi; David S. Urch. Chemical Effects in the X-Ray Emission (F K alpha) and X-Ray Photoelectron Spectra of Alkali-metal and Alkaline-earth-metal Fluorides. *Journal of the Chemical Society, Faraday Transactions*, 1990, Vol. 86, 247-252. DOI: 10.1039/FT9908600247
- 45.** Anna Makarowicz; Christine L. Bailey; Norbert Weiher; Erhard Kemnitz; Sven L. M. Schroeder; Sanghamitra Mukhopadhyay; Adrian Wander; Barry G. Searle; Nicholas M. Harrison. Electronic structure of Lewis acid sites on high surface area aluminium fluorides: a combined XPS and ab initio investigation. *Physical Chemistry Chemical Physics*, 2009, Vol. 11, 5664-5673. DOI: 10.1039/B821484K
- 46.** Victoria Bemmer; Michael Bowker; James H. Carter; Philip R. Davies; Lee E. Edwards; Kenneth D. M. Harris; Colan E. Hughes; Fiona Robinson; David J. Morgan; Matthew G. Thomas. Rationalization of the X-ray photoelectron spectroscopy of aluminium phosphates synthesized from different precursors. *RSC Advances*, 2020, Vol. 10, 8444-8452. DOI: 10.1039/C9RA08738A
- 47.** Sebastian A. Skaavik; Jeffrey D. Henderson; James J. Noël; Mark C. Biesinger. Speciation of Magnesium Surfaces by X-Ray Photoelectron Spectroscopy (XPS). *Surface and Interface Analysis*, 2025, Vol. 57, 717-728. DOI: 10.1002/sia.70003
- 48.** D.A. Hensley; S.H. Garofalini. XPS investigation of lithium borate glass and the Li/LiBO₂ interface. *Applied Surface Science*, 1994, Vol. 81, 331-339. DOI: 10.1016/0169-4332(94)90290-9
- 49.** Kim S. Siow; Leanne Britcher; Sunil Kumar; Hans J. Griesser. XPS Study of Sulfur and Phosphorus Compounds with Different Oxidation States. *Sains Malaysiana*, 2018, Vol. 47, 1913-1922. DOI: 10.17576/jsm-2018-4708-33
- 50.** E. A. Il'inchik; V. V. Volkov; L. N. Mazalov. X-ray photoelectron spectroscopy of boron compounds. *Journal of Structural Chemistry*, 2005, Vol. 46, 523-534. DOI: 10.1007/s10947-006-0133-y
- 51.** E. A. Il'inchik. Standards for X-ray photoelectron spectroscopy of boron compounds. *Journal of Applied Spectroscopy*, 2008, Vol. 75, 883-891. DOI: 10.1007/s10812-009-9116-z
- 52.** Luning Wang; Anjali Menakath; Fudong Han; Yi Wang; Peter Y. Zavalij; Karen J. Gaskell; Oleg Borodin; Dinu Iuga; Steven P. Brown; Chunsheng Wang; Kang Xu; Bryan W. Eichhorn. Identifying the components of the solid-electrolyte interphase in Li-ion batteries. *Nature Chemistry*, 2019, Vol. 11, 789-796. DOI: 10.1038/s41557-019-0304-z
- 53.** Reza Younesi; Maria Hahlin; Fredrik Björefors; Patrik Johansson; Kristina Edström. Li-O₂ Battery Degradation by Lithium Peroxide (Li₂O₂): A Model Study. *Chemistry of Materials*, 2013, Vol. 25, 77-84. DOI: 10.1021/cm303226g
- 54.** David G. Castner; Kenneth Hinds; David W. Grainger. X-ray Photoelectron Spectroscopy Sulfur 2p Study of Organic Thiol and Disulfide Binding Interactions with Gold Surfaces. *Langmuir*, 1996, Vol. 12, 5083-5086. DOI: 10.1021/la960465w
- 55.** Marie-Caroline Bourg; Antonella Badia; R. Bruce Lennox. Gold-Sulfur Bonding in 2D and 3D Self-Assembled Monolayers: XPS Characterization. *The Journal of Physical Chemistry B*, 2000, Vol. 104, 6562-6567. DOI: 10.1021/jp9935337
- 56.** Andika Asyuda; Saunak Das; Michael Zharnikov. Thermal Stability of Alkanethiolate and Aromatic Thiolate Self-Assembled Monolayers on Au(111): An X-ray Photoelectron Spectroscopy

Study. The Journal of Physical Chemistry C, 2021, Vol. 125, 21754-21763. DOI: 10.1021/acs.jpcc.1c06984

57. L  na  c Madec; Jian Xia; R  mi Petibon; Kathlyne J. Nelson; Jon-Paul Sun; Ian G. Hill; Jeff R. Dahn. Effect of Sulfate Electrolyte Additives on LiNi_{1/3}Mn_{1/3}Co_{1/3}O₂/Graphite Pouch Cell Lifetime: Correlation between XPS Surface Studies and Electrochemical Test Results. The Journal of Physical Chemistry C, 2014, Vol. 118, 29608-29622. DOI: 10.1021/jp509731y

58. Laurent Castro; R  mi Dedryv  re; Mohammed El Khalifi; P.-E. Lippens; J  r  me Br  ger; Christiane Tessier; Daniel Gonbeau. The Spin-Polarized Electronic Structure of LiFePO₄ and FePO₄ Evidenced by in-Lab XPS. The Journal of Physical Chemistry C, 2010, Vol. 114, 17995-18000. DOI: 10.1021/jp106631v

59. Dana Pantea; Hans Darmstadt; Serge Kaliaguine; Lydia S  mmchen; Christian Roy. Electrical conductivity of thermal carbon blacks: Influence of surface chemistry. Carbon, 2001, Vol. 39, 1147-1158. DOI: 10.1016/S0008-6223(00)00239-6

60. Dana Pantea; Hans Darmstadt; Serge Kaliaguine; Christian Roy. Electrical conductivity of conductive carbon blacks: influence of surface chemistry and topology. Applied Surface Science, 2003, Vol. 217, 181-193. DOI: 10.1016/S0169-4332(03)00550-6